Synthetic Respondents in Survey Research: An Empirical Test Against a Student Survey

Inna A. Gazieva

Russian Presidential Academy of National Economy and Public Administration (RANEPA), Moscow, Russia
ORCID: 0000-0002-0778-7171 · gazieva-ia@ranepa.ru

Preprint. Not peer reviewed. DOI: 10.5281/zenodo.22939959

> **Author's English version.** This is the author's English version of a preprint written in Russian («Синтетические респонденты в анкетном опросе: эмпирическая проверка на данных опроса студентов», file `PAPER–RU.md` in the same deposit). In case of any discrepancy, the Russian text prevails. Russian-language sources are cited in their original form with English titles in brackets.

Abstract. The aim of this paper is to test empirically the limits of applicability of synthetic respondents, that is, large language models answering survey questions on behalf of "digital twins" constructed from the socio-demographic characteristics of real people. Unlike most published tests, which rely on foreign national surveys, this study uses data from the author's own survey, in which the answer of every human participant is known in advance. The empirical basis is a survey of RANEPA students conducted within the "Scientific Code" («Научный код») programme (2024, N = 1,795, a 39-item questionnaire). For each respondent, a twin was constructed from five attributes taken as a whole from that respondent's row of the data file (sex, level and year of study, campus, broad field of study, ethnicity); the DeepSeek language model re-answered the 36 coded questions of the questionnaire in 64,620 independent calls. The results show that the twin reproduces the central tendency and the direction of demographic differences (the direction of gender differences matched on 85% of items), but systematically narrows the spread of answers (mean standard deviation 0.52 vs. 1.04 for humans), avoids the extreme scale points (36% vs. 50%) and the "hard to say" option, weakens associations between answers to questions from different thematic blocks threefold (mean absolute correlation coefficient 0.10 vs. 0.27), and practically fails to convey the individuality of the respondent (agreement of a person's "own" twin 37.9% vs. 36.5% for a random twin). The novelty of the study lies in distinguishing three levels at which a synthetic respondent behaves in fundamentally different ways: the level of distributions, the level of spread, and the level of associations and individual answers. It is concluded that a synthetic respondent is not a measurement instrument but a means of preparing measurement (pretesting a questionnaire), and a hypothesis is formulated about using discrepancies between human and twin answers to detect socially desirable responding.

Keywords: synthetic respondents, silicon sampling, large language models, survey research, sociological research methodology, social desirability, university students

---

Introduction

Empirical sociology and adjacent fields of applied research are facing a systemic crisis of classical data collection methods. The traditional questionnaire survey, long regarded as the "gold standard", is becoming increasingly vulnerable: fieldwork is getting more expensive, its timelines are stretching,

response rates are declining, and respondent “fatigue” and social desirability effects increasingly distort the resulting social knowledge. There is a natural demand for digital alternatives that would make it possible to measure characteristics of social phenomena faster, at larger scale and at lower cost.

One such alternative is so-called synthetic respondents (also referred to in the literature as a “silicon sample”, AI respondents or virtual respondents), that is, large language models answering survey questions on behalf of a “digital twin” constructed from the demographic description of a real person. Services offering ready-made “algorithmic panels” have already entered the open market, and for a young researcher without a substantial budget the temptation to delegate the collection of opinions to a neural network becomes almost irresistible.

At the same time, in our view, the limits of applicability of such simulations remain a “grey zone”. Their validity is most often discussed on other people’s data and abstract cases, which makes it difficult for a practising researcher to understand where social reality ends and algorithmic illusion begins, which metrics of synthetic data can be trusted, and which hidden distortions make their use scientifically untenable.

The aim of this paper is to test empirically the limits of applicability of synthetic respondents on material from the author’s own survey, in which the answer of every human participant is known in advance. To this end, a synthetic twin is built for each of the 1,795 survey participants, after which the two data sets, human and synthetic, coded by the same rules, are compared at the levels of distributions, spread of answers, associations between answers, and individual respondents.

Literature review

The idea of synthetic respondents goes back to the work of L. Argyle and colleagues (Argyle et al., 2023), which introduced the notions of a “silicon sample” and “algorithmic fidelity”: a language model conditioned on demographic characteristics reproduces part of the aggregate distributions of real surveys. The optimistic line also includes the review by M. Sarstedt and colleagues (Sarstedt et al., 2024), which discusses the opportunities and risks of silicon samples in marketing research, and the work of S. Sun and colleagues (Sun et al., 2024), in which subgroup opinions are generated from group-level demographic data, albeit with the caveat that replicability depends on the group and the topic of the question.

The sceptical line documents three recurring themes. The first is the collapse of variance: model answers become homogenised and are pulled towards the modal option (Li, Li, Qiu, 2025; Dominguez-Olmedo, Hardt, Mendler-Dünner, 2024). The second is the loss of the structure of associations between variables: according to J. Bisbee and colleagues, synthetic answers show less variation than real ones, and regression coefficients estimated on synthetic data often differ significantly from those obtained on survey data (Bisbee et al., 2024); by K. O. Zhulikov’s account, 48% of the coefficients differed, and in 32% of the discrepancies the sign of the effect was reversed (Zhulikov, 2026, p. 152). The third is the under-representation of minorities, which dissolve into the “typical” answer. Importantly, homogenisation is also observed in the largest models of their time: in the study by D. Li and colleagues these are GPT-4 and Llama 3.1 with 405 billion parameters (Li, Li, Qiu, 2025), which suggests that we are dealing with a structural property of the method rather than a growing pain that the next model release will cure. A separate cross-cultural line shows that models are systematically biased towards Western respondents, including on World Values Survey data (Qu, Wang, 2024; AlKhamissi et al., 2024).

The Russian-language literature on the problem is only taking shape. Zh. V. Puzanova and colleagues (Puzanova, Kozhoridze, Kozhoridze, 2025), noting that domestic empirical tests are still few, tested a "virtual respondent" in a qualitative design, in focus groups with young people: the model's answers turned out to be generalised, lacking concrete examples and missing some positions, while the authors themselves attribute the high agreement in their second experiment to the fact that the model was summarising data already obtained from people. K. O. Zhulikov, reviewing publications from 2022–2026, notes that model answers "in repeated tests resembled the answers of one and the same person rather than of different respondents" and considers the use of synthetic respondents justified "at the preparatory stages of work" (Zhulikov, 2026, pp. 155–156). E. E. Kuchko and colleagues propose restricting them to exploratory research and preliminary hypothesis testing, whose results "are subject to mandatory subsequent validation through a survey of real respondents" (Kuchko, Vishnyakova, Marchuk, 2025, p. 33).

Thus, most tests have been carried out on foreign national surveys, while domestic studies are either reviews or qualitative pilots. The author was unable to find in the available literature any quantitative tests on Russian data in which a twin is compared with a specific human respondent; the present study is intended to partly fill this gap.

Data and method

Empirical basis. The benchmark is data from a survey of RANEPA students conducted in 2024 within the "Scientific Code" programme (N = 1,795; a 39-item questionnaire). The data file contains every participant's answer to every question, which allows synthetic answers to be compared not only with aggregate distributions but also with specific people. The primary survey data are not published.

Constructing the twin. For each of the 1,795 respondents, a synthetic twin was built from five attributes: sex, level and year of study, campus of the Academy, broad field of study, and ethnicity. Crucially, the description of each twin was taken as a whole from the row of one real respondent rather than assembled from separate sample shares. If portraits were generated by multiplying marginal distributions (so many per cent women, so many per cent first-year students, so many per cent future lawyers), the sample would inevitably contain "impossible" respondents, for example students enrolled in a field of study not offered at their campus. Borrowing the whole row preserves the joint distribution of attributes exactly as it exists in reality.

Persona prompt. The system part of the request to the model read (translated from Russian): "You are a RANEPA student. What is known about you: {sex}, {year}, {campus}, field of study — {field}, ethnicity — {ethnicity}. Answer the questionnaire as such a student would. Return ONLY the code of the answer option, without explanations." The question was supplied separately, with the full list of options in the form "code — label" and a requirement on the answer format (single choice, rating-scale matrix or multiple choice).

Model and procedure. The DeepSeek language model (version v4-flash, with reasoning mode disabled) was used at a temperature of 0.8, which sets the degree of diversity of answers and leaves the model some freedom without making its answers random. The questionnaire was not presented to the twin as a whole but one block of questions per call, and each call started "from a clean slate": the model had no information about its previous answers. In total, 1,795 × 36 = 64,620 independent calls were made (36 of the 39 questions entered the comparison, since the three open-ended questions were not coded). This is an essential condition: given the whole questionnaire, the model

would start making its answers consistent with one another, and what would be measured is the model's own internal consistency rather than its ability to reproduce people.

Coding. Answers were accepted under a strict rule: only the code of an existing answer option counted; of the 64,620 answers, one did not match the required format. Evasive options ("hard to say", "no experience") were retained as a separate category. The coding of the synthetic data set fully mirrors that of the human one, which is what makes a direct comparison possible.

Comparison measures. The comparison was carried out at four levels: (1) agreement between the answers of a twin and of the corresponding person (exact and within one scale point); (2) spread of answers (mean item-level standard deviation, share of extreme scale values, share of the "hard to say" option); (3) demographic sensitivity (agreement in the direction of differences between subgroups) and associations between answers (mean absolute correlation coefficient between items belonging to different thematic blocks of the questionnaire); (4) the individual level (agreement of a twin with its "own" respondent compared with a random one).

Results

Agreement of answers. The twin answers "close" to the person but rarely in exactly the same way: 37.8% of answers matched exactly, and 79.1% matched within one scale point.

Pull towards the centre. The main defect of synthetic answers is the narrowing of spread. The mean standard deviation of answers is half as large for twins as for humans (0.52 vs. 1.04), and twins choose the extreme scale points noticeably less often (36% vs. 50%). Real opinions are widely distributed, whereas synthetic ones gather in a narrow band around the mean.

Uncertainty. The twin almost never expresses uncertainty. Where human students choose "hard to say" in 13–19% of cases, the model practically never chooses this option. The exception is telling: in questions where the zero code denotes an actual absence of experience ("no experience") rather than difficulty in answering, the model chooses it as often as people do. In other words, the model avoids "I don't know" but not "I haven't done it".

Demographic sensitivity. The demographic portrait does work: the direction of gender differences is reproduced by the twins on 85% of items, that is, the model answers differently for different respondents, broadly reflecting the specified socio-demographic characteristics. At the same time, atypical and small groups are reproduced worse: students in IT fields matched their twins noticeably less often than students in "typical" law and management fields (31.5% vs. 40.1%).

Individual level. The twin does not convey the respondent's individuality: a person's "own" twin resembles that particular person only slightly more than a random twin does (37.9% vs. 36.5%).

Associations between answers. Among human respondents, associations between answers to questions from different thematic blocks are three times stronger than among twins (mean absolute correlation coefficient 0.27 vs. 0.10). This means that the twin loses the interrelation of attitudes and practices characteristic of people, on which the analysis of their structure is based.

A case of social desirability. The most revealing item was the question on the use of ChatGPT: the option "I use it to generate text" was chosen by 27.5% of real students and by 96.8% of their twins. In our view, the model, which is not prone to self-censorship, names the obvious use, whereas human respondents keep silent about it; in this case the discrepancy reveals under-reporting among people rather than an error of the model.

Discussion

The results make it possible to distinguish three levels at which a synthetic respondent behaves in fundamentally different ways.

At the level of distributions and directions, the twin works. The distributions of synthetic answers for most items lie close to the human ones, and the direction of gender differences matches on 85% of items, so to the question "which way is the group leaning?" synthetic data give a reasonable first approximation.

At the level of spread, the twin errs systematically, towards narrowing. It is from spread that the quantities a researcher relies on are calculated: the strength of differences between groups (effect size), confidence intervals, and the sample size needed for a reliable result. Since the twins' answers are packed more tightly than those of people, all these quantities will be biased when estimated on synthetic data. Importantly, the bias is not random but systematic and therefore cannot be removed by increasing the number of twins: however many there are, they will still answer more consistently than people do.

At the level of associations and the individual person, the twin does not work. Its associations between answers to different questions are three times weaker, and cases in which an association reverses its sign on synthetic data have been described in the literature (Bisbee et al., 2024; Zhulikov, 2026, p. 152). In the author's view, it is impossible to reconstruct the structure of value orientations from such material (on the methodology of measuring it, see: Gazieva, 2024) or to predict the answer of a specific person.

Our data are consistent with the sceptical line of the literature and refine it on Russian student material. They also make it possible to refine the recommendation of Zh. V. Puzanova and colleagues to begin with a pilot survey of real people and only then build a generalised virtual respondent (Puzanova, Kozhoridze, Kozhoridze, 2025, p. 233): we fully agree with the first part of this recommendation, but the generalised respondent, in the authors' own words, serves "to represent the typical position of the participants", and the typical position is precisely what synthetic data pull the spread towards.

Practical application. It follows that a synthetic twin is not a measurement instrument but a means of preparing measurement. It is useful exactly up to the moment when the researcher starts obtaining figures from it that are intended for the text of their work: from that moment on it measures not people's opinions but the behaviour of a language model. Justified uses include: running a draft questionnaire to detect poorly worded items (if the model systematically breaks the answer format on an item, or all twins choose the same option, this most likely indicates a defect in the wording rather than unanimity); comparing two versions of a question on the same grid of personas; and obtaining a rough idea of the set of answer options that may be chosen at all, but not of their distribution.

The requirements for persona descriptions depend on the researcher's claims. For finding poorly worded items, representativeness is not essential: a few dozen personas covering the "edges of the population" are enough. Claims about distributions require a joint socio-demographic structure, that is, row-level information about individual people rather than summary percentages. Possible sources are administrative data of one's own organisation, data from earlier studies of the same group (with all their inherent biases), open survey data sets, and the socio-demographic block of one's own pilot survey.

Hypothesis: the language model as a mirror of self-censorship. The discrepancy on the ChatGPT question suggests a possible methodological technique. If a small survey of real people is conducted and its results are compared with the twins' answers, questions with especially large discrepancies may point to items where people under-report or embellish their answers. Such items could then be moved to indirect techniques, for example a list experiment. We stress that this has not been tested as a method: we have a single observation on a single question. The procedure has not been run on a questionnaire with sensitive items known in advance, the false-positive rate has not been assessed, and distinguishing a respondent's self-censorship from the model's ignorance of Russian realities remains to be done. Until such a test, this is a hypothesis for independent research rather than a recommendation.

Limitations

A single language model was tested on a single questionnaire and a single population (students of one university), so the conclusions apply to this model and this instrument; comparison with other models is a natural continuation of the work. The generation temperature of 0.8 is a design choice, and the aggregate distributions depend on it. Each twin was surveyed once, which allows the representativeness of the aggregate to be assessed but not the stability of an individual twin's answers. Grouping fields of study into broad categories coarsens the corresponding attribute. Finally, the twin knew only five socio-demographic attributes of the person; richer persona descriptions may yield different results at the individual level, although, according to the literature, they do not eliminate the narrowing of spread.

Conclusion

The test showed that a synthetic respondent based on a large language model is a "smoothed shadow" of the real sample: it guesses the direction in which a conclusion is drawn but systematically loses the spread, the uncertainty, the associations between answers and the individuality of the respondent. This makes it possible to formulate a principle that, in our view, applies to the use of artificial intelligence tools in research in general: a neural network speeds up preparatory work but substitutes neither for the measurement itself nor for the verification of its results. What is reasonable is not replacing people with a model but a hybrid scheme in which the twin speeds up the preparation of the instrument, while an obligatory human minimum remains a condition of measurement.

Acknowledgements

The author thanks Ruslan Gaziev for the technical implementation of the experiment: the software and the work with the language model.

References

1. Gazieva I. A. *Sotsiologiya tsennostei: metodologiya issledovaniya* [Sociology of Values: Research Methodology]: monograph. Moscow: INFRA-M, 2024. 253 p. DOI: 10.12737/2133680. (In Russian)
2. Zhulikov K. O. Kriterii otsenki sinteticheskikh respondentov v kachestvennykh issledovaniyakh: granitsy primenimosti [Criteria for Evaluating Synthetic Respondents in Qualitative Research: Limits of Applicability]. *Interaktsiya. Interv'yu. Interpretatsiya*

[Interaction. Interview. Interpretation]. 2026. Vol. 18, No. 2. P. 147–160. DOI: 10.19181/inter.2026.18.2.5. (In Russian)
3. Kuchko E. E., Vishnyakova M. V., Marchuk Ya. I. Opros sinteticheskikh respondentov: novye vozmozhnosti ili ugroza dostovernosti dannykh? [Survey of Synthetic Respondents: New Opportunities or a Threat to Data Reliability?]. *Zhurnal Belorusskogo gosudarstvennogo universiteta. Sotsiologiya* [Journal of the Belarusian State University. Sociology]. 2025. No. 3. P. 31–35. URL: https://elib.bsu.by/bitstream/123456789/350054/1/31-35.pdf. (In Russian)
4. Puzanova Zh. V., Kozhoridze G. G., Kozhoridze D. G. II i sotsiologiya: analiz tekhnologicheskikh vozmozhnostei virtual'nykh respondentov [AI and Sociology: Analysis of the Technological Capabilities of Virtual Respondents]. *Sotsiologiya: metodologiya, metody, matematicheskoe modelirovanie (Sotsiologiya: 4M)* [Sociology: Methodology, Methods, Mathematical Modeling (Sociology: 4M)]. 2025. No. 60. P. 216–246. DOI: 10.19181/4m.2025.34.1.6. (In Russian)
5. AlKhamissi B., ElNokrashy M., AlKhamissi M., Diab M. Investigating Cultural Alignment of Large Language Models. In: *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL 2024)*. 2024. arXiv: 2402.13231.
6. Argyle L. P., Busby E. C., Fulda N., Gubler J. R., Rytting C., Wingate D. Out of One, Many: Using Language Models to Simulate Human Samples. *Political Analysis*. 2023. Vol. 31, No. 3. P. 337–351. DOI: 10.1017/pan.2023.2.
7. Bisbee J., Clinton J. D., Dorff C., Kenkel B., Larson J. M. Synthetic Replacements for Human Survey Data? The Perils of Large Language Models. *Political Analysis*. 2024. Vol. 32, No. 4. P. 401–416. DOI: 10.1017/pan.2024.5.
8. Dominguez-Olmedo R., Hardt M., Mendler-Dünner C. Questioning the Survey Responses of Large Language Models. In: *Advances in Neural Information Processing Systems (NeurIPS 2024)*. 2024. arXiv: 2306.07951.
9. Li D., Li L., Qiu H. S. ChatGPT is not A Man but Das Man: Representativeness and Structural Consistency of Silicon Samples Generated by Large Language Models. 2025. arXiv: 2507.02919.
10. Qu Y., Wang J. Performance and biases of Large Language Models in public opinion simulation. *Humanities and Social Sciences Communications*. 2024. Vol. 11. Art. 1095. DOI: 10.1057/s41599-024-03609-x.
11. Sarstedt M., Adler S. J., Rau L., Schmitt B. Using large language models to generate silicon samples in consumer and marketing research: Challenges, opportunities, and guidelines. *Psychology & Marketing*. 2024. Vol. 41, No. 6. P. 1254–1270. DOI: 10.1002/mar.21982.
12. Sun S., Lee E., Nan D., Zhao X., Lee W., Jansen B. J., Kim J. H. Random Silicon Sampling: Simulating Human Sub-Population Opinion Using a Large Language Model Based on Group-Level Demographic Information. 2024. arXiv: 2402.18144.